MATH INQUIRERS HOLIDAY PROGRAM

Readiness Tracker and Puzzle Set

Tutor notes on how each child thinks: four signs to track and an 8-question Go further puzzle set

Notice it. Test it. Explain why.

How is each child thinking? We look at how a child thinks, not only the score. These notes are for tutors only: never show them to children or share them between families. Tick after each lesson (30 seconds per child).

The four signs

Code	Sign	What you see or hear
A	Solved a Go further question alone	Got the answer without hints, and showed how (counters, drawing or words).
B	Found a second way	Used a different method or found another answer, and said how it is different.
C	Noticed a pattern and said why	Said 'it always goes up by 3 because...', not just 'it goes up'.
D	Stretched it to a new case	Said what would happen with a bigger number or a changed rule, and checked it.

Lesson log: copy one page per lesson

Lesson number: _____ Date: _____ Tutor: _____

Child	A	B	C	D	Total	Note (what they said or did)
1.						
2.						
3.						
4.						
5.						
6.						

7.						
8.						

Programme summary: one row per child

Write each lesson's total (0 to 4) in the box. Then add up in the last columns.

Child	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Lessons with 3+	Decision
1.												
2.												
3.												
4.												
5.												
6.												
7.												
8.												

How to decide

- 3 or 4 in at least 3 lessons: remind the child that Go further and the Investigation are theirs to choose, and invite the whole group to a short challenge session open to all. The Go further puzzle set below is for any child who would like it.
- 2 in most lessons: keep going and give more Think Harder questions in class.
- Mostly 0 or 1: do not worry. Check understanding of the core questions first. Many children grow later in the programme.
- One lesson alone means little. Look at the pattern across lessons, and at what the child actually says.
- Use this only to decide what to offer next, never to label or group a child. Do not share rankings between families.

Go further puzzle set: child sheet

Name: _____ Date: _____

Take your time. For each question, show how you found your answer with a drawing, counters or words. Then tell your tutor why.

1. A pattern adds a number bigger than 0 each time, its 5th number is 20, and it starts at a whole number bigger than 0. How many different patterns can you make? List the first number and the rule for each.

_____

2. 12 children voted for three fruits. No two fruits got the same number of votes. The most votes was 5. How many votes did each fruit get?

_____

3. Squares in a row use 4, 7, 10 … sticks. Mei says she can make a row of squares using EXACTLY 30 sticks. Can she? What about exactly 31 sticks? Explain how you know.

_____

4. Zoe is 6th from the front of a queue and 6th from the back. How many children are in the queue? What if a child is 10th from the front and 10th from the back? Can you find a quick way?

_____

5. You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.

_____

6. A big triangle has lines drawn from its top corner to the bottom. With 1 line there are 3 triangles, with 2 lines 6, with 3 lines 10. How many triangles with 4 lines? With 5 lines? Draw it to check.

_____

7. Ali, Ben and Mei stand in a line. List every different order they can stand in. How many are there? How do you know you have them all? If Ben must NOT be first, how many orders are left?

_____

8. Triangles in a row use 3, 5, 7 … sticks. Squares in a row use 4, 7, 10 … sticks. Which numbers of sticks, up to 31, can make a row of triangles AND (with the same number of sticks) a row of squares, with none left over? What do you notice?

_____

Go further puzzle set: tutor guide and answer key

- Use one to one or in a group of up to 4. Allow about 25 minutes. Read each question aloud if needed.
- Do not give hints in the first attempt. After 2 minutes you may give one hint, and score it as 'with a hint'.
- Score each question: 2 = correct and explains why (without hint); 1 = correct, or right idea with a hint; 0 = not yet.
- Then ask the extension question once. A good answer earns a star. Stars show how far the child can stretch.
- Total out of 16. As a guide, 12 or more with 3 or more stars suggests a child who would enjoy more open puzzles. Use it only to choose what to offer next, never to label a child. Combine it with the lesson tracker; neither alone decides.
- These problems practise patterns, listing in order, working backwards and logic. This set is not an official test and is not linked to any school or MOE programme.

#	Question	Answer	Extension (star)

1	A pattern adds a number bigger than 0 each time, its 5th number is 20, and it starts at a whole number bigger than 0. How many different patterns can you make? List the first number and the rule for each.	4 patterns: start 16 add 1; start 12 add 2; start 8 add 3; start 4 add 4. (Start 0 or a jump of 5 would need a first number of 0.)	What if the 5th number was 30? How many patterns now? How do you know you have them all?
2	12 children voted for three fruits. No two fruits got the same number of votes. The most votes was 5. How many votes did each fruit get?	5, 4 and 3.	What if there were 15 children and the most votes was 6? Can the votes still all be different?
3	Squares in a row use 4, 7, 10 … sticks. Mei says she can make a row of squares using EXACTLY 30 sticks. Can she? What about exactly 31 sticks? Explain how you know.	No for 30: the stick numbers go 4, 7, 10, 13, 16, 19, 22, 25, 28, 31 and skip 30 (each number is 1 more than a number we say counting in 3s, and 29 is not one). Yes for 31: that is 10 squares.	What about exactly 40 sticks? How do you know before you draw?
4	Zoe is 6th from the front of a queue and 6th from the back. How many children are in the queue? What if a child is 10th from the front and 10th from the back? Can you find a quick way?	11 and 19. Zoe is counted twice if we add 6 + 6, so take away 1. Quick way: double the place number and take away 1 (6 + 6 − 1 = 11; 10 + 10 − 1 = 19).	Zoe is 15th from the front and 15th from the back. How many children? Explain your quick way.
5	You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.	7 totals: 5, 10, 15, 20, 25, 30, 35.	What if you also had a 50¢ coin? How many totals now?
6	A big triangle has lines drawn from its top corner to the bottom. With 1 line there are 3 triangles, with 2 lines 6, with 3 lines 10. How many triangles with 4 lines? With 5 lines? Draw it to check.	15 triangles with 4 lines and 21 with 5 lines. The number added grows by 1 each time: 3, 6, 10, 15, 21 (+3, +4, +5, +6). Check by drawing: count the small triangles first, then the bigger ones made of 2, 3 … small ones.	How many triangles with 6 lines? What do you think the pattern is?
7	Ali, Ben and Mei stand in a line. List every different order they can stand in. How many are there? How do you know you have them all? If Ben must NOT be first, how many orders are left?	6 orders: ABM, AMB, BAM, BMA, MAB, MBA (Ali, Ben, Mei). Fix who is first, then list the two ways for the other two. If Ben is not first: 4 orders.	Four friends: Ali, Ben, Mei and Sam. Sam must be last. How many orders?

8	Triangles in a row use 3, 5, 7 … sticks. Squares in a row use 4, 7, 10 … sticks. Which numbers of sticks, up to 31, can make a row of triangles AND (with the same number of sticks) a row of squares, with none left over? What do you notice?	7, 13, 19, 25 and 31. They go up by 6 (the 2 and the 3 from the two patterns meet every 6). For example 7 sticks make 3 triangles or 2 squares.	The next number of sticks after 31 that works? How do you know?